

MAT 324 HW03

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Problem 1. Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, and $f: X \rightarrow Y$ a map.

- (a) Suppose f is continuous at every $x \in A$. Show that $f|_A$ is continuous.
- (b) Given an example showing that the converse need not hold, i.e. $f|_A$ may be continuous despite f being discontinuous at some $x \in A$.

Solution. (a) If f is continuous on X then, for every open $V \subseteq Y$, $f^{-1}(V)$ is open in X , so $f|_A^{-1}(V) = A \cap f^{-1}(V)$ is open in A ; this is what it means for $f|_A$ to be continuous.

- (b) Take $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

and let $A = [0, \infty)$. Then, $f|_A$ is constant, hence continuous, but f is not continuous at $0 \in A$ because $f(0) \in (\frac{1}{2}, \infty)$ but $f^{-1}((\frac{1}{2}, \infty)) = [0, \infty)$ does not contain a neighborhood of 0. $\square$

Problem 2. Suppose $A \subseteq \mathbb{R}$ is closed and $f: A \rightarrow \mathbb{R}$ is a continuous function. Thus, $\mathbb{R} \setminus A$ is a union of at most countably many disjoint intervals (a_k, b_k) with $a_k, b_k \in A \cup \{\pm\infty\}$ distinct for $k \in \mathbb{N}$. Show that the function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(x) = \begin{cases} f(x) & \text{if } x \in A \\ f(b_k) & \text{if } x \in (a_k, b_k) \text{ for some } k \text{ such that } a_k = -\infty \\ f(a_k) & \text{if } x \in (a_k, b_k) \text{ for some } k \text{ such that } b_k = \infty \\ tf(a_k) + (1-t)f(b_k) & \text{if } x = ta_k + (1-t)b_k \text{ for some } t \in [0, 1] \text{ and } k \end{cases}$$

is continuous.

Solution. If x is not a boundary point of A (i.e. it either lies outside A or in the interior of A) then g is continuous at x because g is continuous on the open sets $\text{Int}(A)$ and $\mathbb{R} \setminus A$.

Now suppose $x \in \partial A$ is on the boundary. We consider two cases. First, if there is no sequence of b_k s converging to x , then there is a neighborhood $B = (x - \delta, x + \delta)$ of x not containing any boundary points besides possibly x . By construction g must be continuous on $(x - \delta, x]$ and $[x, x + \delta)$, since $g \equiv f$ on one of these intervals and g is described by a linear polynomial on the other (because one of the intervals is contained in A and the other isn't); the pasting lemma then gives continuity on the neighborhood B , so g is continuous at x .

The alternative is that some sequence of b_k s converges to x ; call this sequence $(\hat{b}_n)_{n \in \mathbb{N}}$ and let $\hat{a}_n$ be the left endpoint of the interval in $\mathbb{R} \setminus A$ that has $\hat{b}_n$ as its right endpoint. Assume that the $\hat{b}_n$ s are all less than x ; we will show that

$$\lim_{x' \rightarrow x^-} g(x') = g(x)$$

The sequence $\hat{a}_n$ also converges to x because

$$\sup_{\hat{b}_k \leq \hat{a}_n} \hat{b}_k \leq \hat{a}_n \leq \hat{b}_n.$$

Now we have for any $\delta > 0$ that if $x' \in (x - \delta, x] \setminus A$, then

$$|g(x) - g(x')| \leq \max \left(\sup_{\hat{a}_k \in (x - \delta, x]} |g(x) - g(\hat{a}_k)|, \sup_{\hat{b}_k \in (x - \delta, x]} |g(x) - g(\hat{b}_k)| \right)$$

because for each $[\hat{a}_k, \hat{b}_k] \subseteq (x - \delta, x]$, g reaches its maximum and minimum value at the endpoints a_k, b_k . Since the $\hat{a}_k$ and $\hat{b}_k$ converge to x , given $\varepsilon > 0$ we can pick δ so that the upper bound is less than ε , which implies that $|g(x) - g(x')| < \varepsilon$. By continuity of f at x , we can also shrink δ to ensure that if $x' \in (x - \delta, x] \cap A$ then $|g(x) - g(x')| < \varepsilon$ still holds, hence the bound holds for all $x' \in (x - \delta, x]$. Thus

$$\lim_{x' \rightarrow x^-} g(x') = g(x).$$

To get the limit from above, we can find a sequence of a_k s, say $\tilde{a}_n$, (and the corresponding $\tilde{b}_n$) whose terms are all at least as large as x that converges to x . Then, consider $h(x) = g(-x)$, which is now defined as an extension of $x \mapsto f(-x)$ by linear polynomials on intervals $(-b_k, -a_k)$. The $-\tilde{a}_n$ and $-\tilde{b}_n$ converge to $-x$ from below and $-x$ is in the closed set $-A = \{-a : a \in A\}$ on which $x \mapsto f(-x)$ is defined, so the preceding paragraph applies to h and we may use continuity from below for h at $-x$ to extract continuity from above for g at x . Thus g is continuous at all $x \in A$, completing the proof. $\square$

Problem 3. Let $A \subseteq \mathbb{R}$. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x & \text{if } x \in A \\ -x & \text{if } x \notin A \end{cases}$$

is Lebesgue measurable if and only if A is a Lebesgue measurable set.

Solution. Since $\{(a, \infty) : a \in \mathbb{R}\}$ generates the Borel σ -algebra on $\mathbb{R}$, f is Lebesgue measurable if and only if $f^{-1}((a, \infty))$ is Lebesgue for all $a \in \mathbb{R}$. To show that this is equivalent to A being Lebesgue, we write

$$f^{-1}((a, \infty)) = (A \cap (a, \infty)) \cup (A^c \cap (-\infty, -a)).$$

If A is Lebesgue, then this set is of course Lebesgue for all real a because Lebesgue sets form a σ -algebra. Conversely, if $f^{-1}((a, \infty))$ is Lebesgue for all real a , then

$$A \cap (0, \infty) = (0, \infty) \cap f^{-1}((0, \infty)), \quad A \cap (-\infty, 0) = (-\infty, 0) \cap f^{-1}((0, \infty)),$$

are both Lebesgue, implying

$$A \setminus \{0\} = (A \cap (0, \infty)) \cup ((-\infty, 0) \setminus (A^c \cap (-\infty, 0)))$$

is Lebesgue. If $A \setminus \{0\} = A$ then A is Lebesgue, otherwise $(A \setminus \{0\}) \cup \{0\} = A$ so A is still Lebesgue. Thus A is Lebesgue if and only if the pre-image of every open ray under f is Lebesgue. $\square$

Problem 4. Let $(X, \mathcal{S})$ be a measurable space.

- (a) Suppose (Y, d) is a metric space, $g: Y \rightarrow \mathbb{R}$ is a continuous function, and $f: X \rightarrow Y$ is a function such that $f^{-1}(V) \in \mathcal{S}$ for every open subset $V \subseteq Y$. Show that the function $g \circ f: X \rightarrow \mathbb{R}$ is $\mathcal{S}$ -measurable.
- (b) Suppose $f_1, \dots, f_k: X \rightarrow \mathbb{R}$ are $\mathcal{S}$ -measurable functions and $f = (f_1, \dots, f_k): X \rightarrow \mathbb{R}^k$. Show that $f^{-1}(V) \in \mathcal{S}$ for every open subset $V \subseteq \mathbb{R}^k$.
- (c) Show that $f_1, f_2: X \rightarrow \mathbb{R}$ are $\mathcal{S}$ -measurable functions and $a_1, a_2 \in \mathbb{R}$. Show that

$$a_1 f_1 + a_2 f_2: X \rightarrow \mathbb{R} \text{ and } f_1 f_2: X \rightarrow \mathbb{R}$$

are also $\mathcal{S}$ -measurable functions. If in addition $f_2^{-1}(0) = \emptyset$, show that $f_1/f_2: X \rightarrow \mathbb{R}$ is an $\mathcal{S}$ -measurable function.

- (d) Show that a function $f: X \rightarrow \mathbb{R}$ is Lebesgue measurable if and only if the function f^3 is Lebesgue measurable.
- (e) Give an example of a function $f: \mathbb{R} \rightarrow \mathbb{R}$ so that it is not Lebesgue measurable, but $f^2: \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue measurable.

Solution. (a) It suffices to check that the pre-image of an open set under $g \circ f$ is in $\mathcal{S}$. Suppose $U \subseteq \mathbb{R}$ is open. Then $g^{-1}(U)$ is open in Y by continuity, so $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ is in $\mathcal{S}$.

(b) The rational cubes $(a_1, b_1) \times \dots \times (a_k, b_k)$, where $a_j < b_j$ for $j = 1, \dots, k$ are rational numbers, are a basis for the topology of $\mathbb{R}^k$. That is, for any open $V \subseteq \mathbb{R}^k$, we can find cubes $I_n = (a_1^{(n)}, b_1^{(n)}) \times \dots \times (a_k^{(n)}, b_k^{(n)})$, where the intervals have endpoints in $\mathbb{Q}$, such that $V = \bigcup_{n=1}^{\infty} I_n$. Then,

$$f^{-1}(V) = \bigcup_{n=1}^{\infty} f^{-1}(I_n) = \bigcup_{n=1}^{\infty} \bigcap_{j=1}^k f_j^{-1}((a_j^{(n)}, b_j^{(n)}))$$

because the pre-image of I_n is the set of $x \in X$ such that $f_j(x) \in (a_j^{(n)}, b_j^{(n)})$ for each $j = 1, \dots, k$. In summary, $f^{-1}(V)$ is the union of countably many finite intersections of $\mathcal{S}$ -measurable sets, since of course $f_j^{-1}((a_j^{(n)}, b_j^{(n)}))$ is $\mathcal{S}$ -measurable, so $f^{-1}(V)$ is $\mathcal{S}$ -measurable.

(c) Define $f: X \rightarrow \mathbb{R}^2$ by $f(x) = (f_1(x), f_2(x))$; this is $\mathcal{S}$ -measurable because, by part (b), the pre-image of every open subset of $\mathbb{R}^2$ under f is in $\mathcal{S}$, which by part (a) implies that f is a $\mathcal{S}$ -measurable function. Then, the functions $g_1, g_2: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g_3: \mathbb{R} \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ given by

$$g_1(x, y) = a_1 x + a_2 y, \quad g_2(x, y) = xy, \quad g_3(x, y) = x/y$$

are all continuous on their domains, so by (a) the functions $g_1 \circ f = a_1 f_1 + a_2 f_2$ and $g_2 \circ f = f_1 f_2$ are $\mathcal{S}$ -measurable, and if $f_2^{-1}(0) = \emptyset$, then $g_3 \circ f = f_1/f_2$ is also $\mathcal{S}$ -measurable.

(d) If $f: X \rightarrow \mathbb{R}$ is measurable then we can compose f with the continuous map $x \mapsto x^3$ to get that f^3 is also measurable by (a). Conversely if $f^3: X \rightarrow \mathbb{R}$ is measurable then we compose f^3 with the continuous map $x \mapsto \sqrt[3]{x}$ to get that $\sqrt[3]{f^3} = f$ is also measurable by (a).

(e) Let $V \subseteq \mathbb{R}$ be a non-Lebesgue set. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in V \\ -1 & \text{if } x \notin V. \end{cases}$$

Then the pre-image of the Borel set $\{1\}$ is V , so f is not Lebesgue measurable, but $f^2 \equiv 1$ everywhere so obviously it is Lebesgue measurable. $\square$

Problem 5. Let $(X, \mathcal{S}, \mu)$ be a measurable space and $f_1, f_2, \dots : X \rightarrow [-\infty, \infty]$ be $\mathcal{S}$ -measurable functions converging to a function $f : X \rightarrow [-\infty, \infty]$ pointwise. Thus, f is also $\mathcal{S}$ -measurable and the subsets $X^\pm = f^{-1}(\pm\infty)$ are elements of $\mathcal{S}$.

- (a) Suppose $\mu(X^+) < \infty$. Show that for every $\varepsilon > 0$ there exists $X_\varepsilon^+ \in \mathcal{S}$ so that $X_\varepsilon^+ \subseteq X^+$, $\mu(X^+ \setminus X_\varepsilon^+) < \varepsilon$, and the functions $f_1|_{X_\varepsilon^+}, f_2|_{X_\varepsilon^+}, \dots$ converge uniformly to ∞ .
- (b) Suppose $\mu(X) < \infty$. Show that for every $\varepsilon > 0$ there exists $X_\varepsilon \in \mathcal{S}$ so that $\mu(X \setminus X_\varepsilon) < \varepsilon$ and the functions $f_1|_{X_\varepsilon}, f_2|_{X_\varepsilon}, \dots$ converge uniformly to $f|_{X_\varepsilon}$.
- (c) Give an example showing that the last conclusion can fail if $\mu(X) = \infty$.

Solution. (a) For $N, t \in \mathbb{N}$, let $X_{N,t} = X^+ \cap \bigcap_{n=N}^\infty f_n^{-1}([t, \infty])$; this is the subset of X^+ on which $f_n(x) \geq t$ for all $n \geq N$. Then,

$$X^+ = \bigcup_{N=1}^\infty X_{N,t}$$

because for each $x \in X^+$, given $t \in \mathbb{N}$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $f_n(x) \geq t$ — this is what pointwise convergence means. Clearly $X_{N,t} \subseteq X_{N+1,t}$ for all N and t , so monotone convergence of measure gives

$$\mu(X^+) = \lim_{N \rightarrow \infty} \mu(X^+ \cap X_{N,t})$$

for all $t > 0$. The limit is finite, so for each positive integer t there is an N_t such that $\mu(X^+)$ is within $\varepsilon/2^t$ of $\mu(X_{N_t,t})$, hence $\mu(X^+ \setminus X_{N_t,t}) < \varepsilon/2^t$. Let

$$X_\varepsilon^+ = \bigcap_{t=1}^\infty X_{N_t,t}.$$

For each $t \in \mathbb{N}$, $X_\varepsilon^+ \subseteq X_{N_t,t}$, which means that there is an $N = N_t \in \mathbb{N}$ depending only on t such that $f_n(x) \geq t$ for all $n \geq N$, i.e. convergence of $f_1, f_2, \dots$ on X_ε^+ is uniform. Furthermore,

$$\mu(X^+ \setminus X_\varepsilon^+) \leq \mu\left(\bigcup_{t=1}^\infty X^+ \setminus X_{N_t,t}\right) \leq \sum_{t=1}^\infty \mu(X^+ \setminus X_{N_t,t}) < \sum_{t=1}^\infty \varepsilon/2^t = \varepsilon,$$

so X_ε^+ has the desired properties.

(b) Let $X^0 = f^{-1}(\mathbb{R}) \in \mathcal{S}$ be the set of points on which f is finite. By Egorov's theorem for sequences of functions that converge to a function mapping into the reals, there exists $X_{\varepsilon/3}^0 \subseteq X^0$ in $\mathcal{S}$ such that $\mu(X^0 \setminus X_{\varepsilon/3}^0) < \varepsilon/3$ and $f_1|_{X_{\varepsilon/3}^0}, f_2|_{X_{\varepsilon/3}^0}, \dots$ converges uniformly to $f|_{X_{\varepsilon/3}^0}$.

By part (a), there exists $X_{\varepsilon/3}^+ \subseteq X^+$ in $\mathcal{S}$ such that $\mu(X^+ \setminus X_{\varepsilon/3}^+) < \varepsilon/3$ and the sequence $f_1|_{X_{\varepsilon/3}^+}, f_2|_{X_{\varepsilon/3}^+}, \dots$ converges uniformly to $f|_{X_{\varepsilon/3}^+}$.

Similarly, applying (a) to the sequence $-f_1|_{X^-}, -f_2|_{X^-}, \dots$ converging to $-f|_{X^-} \equiv \infty$ yields the existence of a set $X_{\varepsilon/3}^- \subseteq X^-$ in $\mathcal{S}$ such that $\mu(X^- \setminus X_{\varepsilon/3}^-) < \varepsilon/3$ and the sequence $-f_1|_{X_{\varepsilon/3}^-}, -f_2|_{X_{\varepsilon/3}^-}, \dots$ converges uniformly to $-f|_{X_{\varepsilon/3}^-}$, or equivalently $f_1|_{X_{\varepsilon/3}^-}, f_2|_{X_{\varepsilon/3}^-}, \dots$ converges uniformly to $f|_{X_{\varepsilon/3}^-}$.

Then, take $X_\varepsilon = X_{\varepsilon/3}^0 \cup X_{\varepsilon/3}^+ \cup X_{\varepsilon/3}^-$, so that

$$\mu(X \setminus X_\varepsilon) \leq \mu(X \setminus X_{\varepsilon/3}^0) + \mu(X \setminus X_{\varepsilon/3}^+) + \mu(X \setminus X_{\varepsilon/3}^-) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

and we have uniform convergence on each of X_ε due to having uniform convergence on the component sets $X_{\varepsilon/3}^0, X_{\varepsilon/3}^+, X_{\varepsilon/3}^-$.

(c) Let $f_1, f_2, \dots : \mathbb{R} \rightarrow \mathbb{R}$ be a sequence of functions where f_n is the indicator function $\mathbf{1}_{[n, n+1]}$. Then, f_n converges pointwise to the zero function f , but, if $\varepsilon < 1$, then any X_ε such that $|\mathbb{R} \setminus X_\varepsilon| < \varepsilon$ intersects each $[n, n+1]$ interval (otherwise $\mathbb{R} \setminus X_\varepsilon$ contains an interval of length 1, hence has outer measure bigger than ε), so for any $n \in \mathbb{N}$ we have

$$\sup_{x \in X_\varepsilon} |f(x) - f_n(x)| = 1,$$

meaning the convergence to f is not uniform. $\square$

Problem 6. Let $\mathbb{Q} = \{q_1, q_2, \dots\}$, $I_k = (q_k - 2^{-k}, q_k + 2^{-k})$ for $k \in \mathbb{N}$, $U = \bigcup_{k=1}^{\infty} I_k$, and $A \subseteq \mathbb{R}$.

(a) Suppose $|\mathbb{R} \setminus A| = 0$. Show that $A \cap I_k \neq \emptyset$ for every $k \in \mathbb{N}$ and $A \cap U$ is dense in A .

(b) Suppose $\mathbf{1}_U|_A$ is continuous. Show that $|\mathbb{R} \setminus A| > 0$.

Solution. (a) We show the contrapositive: if $A \cap I_k = \emptyset$ for some $k \in \mathbb{N}$, then $I_k \subseteq \mathbb{R} \setminus A$, so $|\mathbb{R} \setminus A| \geq |I_k| = 2 \cdot 2^{-k}$ is nonzero.

Suppose that V is a nonempty open subset of A , so $V = V' \cap A$ for some nonempty open subset V' of $\mathbb{R}$. Then, for each $x \in V'$, there is an $\varepsilon > 0$ such that the open ball $B_\varepsilon(x)$ centered at x is a subset of V' . Choose $N \in \mathbb{N}$ such that $2^{-N} < \varepsilon/2$, so that $B_{2^{-N}}(x) \subseteq B_\varepsilon(x)$. Since $B_{2^{-N}}(x)$ contains infinitely many rational numbers, there exists an $n \geq N$ such that $q_n \in B_{2^{-N}}(x)$. In particular, any point $y \in B_{2^{-N}}(q_n) = I_n$ satisfies

$$|x - y| \leq |x - q_n| + |q_n - y| < 2^{-N} + 2^{-n} \leq 2 \cdot 2^{-N} < \varepsilon,$$

so V' contains I_n . Since $A \cap I_n$ is nonempty, A contains a point of $U \cap V' \supseteq I_n$, so $U \cap A \cap V' = U \cap A \cap V$ is nonempty, which is what it means for $U \cap A$ to be dense in A .

(b) Suppose for the sake of contradiction that $|\mathbb{R} \setminus A| = 0$, so $|A|$ is infinite (else $|A \cup (\mathbb{R} \setminus A)| = |\mathbb{R}|$ would be finite). The indicator function of U restricted to A is the same thing as the function $\mathbf{1}_{U \cap A} : A \rightarrow \mathbb{R}$. Since $\mathbf{1}_{U \cap A}$ is constant on $U \cap A$, it is uniformly continuous on $U \cap A$, so, since $U \cap A$ is dense in A by part (a) (and our assumption that $|\mathbb{R} \setminus A| = 0$), $\mathbf{1}_{U \cap A}$ uniquely extends to a (uniformly) continuous function on A ; this is an easy result from MAT 320.

One continuous function $A \rightarrow \mathbb{R}$ whose restriction to $U \cap A$ is $\mathbf{1}_{U \cap A}$ is the constant function $\mathbf{1}_A : A \rightarrow \mathbb{R}$. By uniqueness it must be the only such function, implying $\mathbf{1}_U|_A$ is the same function as $\mathbf{1}_A|_A$, which forces $U \cap A = A$, so $A \subseteq U$. However,

$$|U| \leq \sum_{k=1}^{\infty} |I_k| = \sum_{k=1}^{\infty} 2 \cdot 2^{-k} = 2,$$

so $|A|$ is finite because A is contained in a set of finite measure, which is a contradiction. $\square$